Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 PHYSICS

9478/03

15 September 2026

2 hours

Paper 3 Longer Structured Questions

You must answer on the question paper.
No additional materials are needed.

INSTRUCTIONS

- Section A: answer **all** questions.
- Section B: answer **one** question.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and index number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use an approved calculator.

- **INFORMATION**
- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

For Examiner's Use	
Section A	
1	/10
2	/8
3	/12
4	/9
5	/9
6	/7
Section B	
7	/20
8	/20
s.f.	/−1
Total	/75

This document has **36** pages. Any blank pages are indicated.

Data

speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}\,\text{H m}^{-1}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$
	$(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1})$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
unified atomic mass constant	$u = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31}\,\text{kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27}\,\text{kg}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Avogadro constant	$N_A = 6.02 \times 10^{23}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T/\text{K} = T/^\circ\text{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

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displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + ...$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + ...$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + ...$
capacitors in parallel	$C = C_1 + C_2 + ...$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0 e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Section A

Answer **all** the questions in this Section in the spaces provided.

1 A battery of electromotive force (e.m.f.) 9.6 V and negligible internal resistance is connected in series with two fixed resistors and a thermistor, as shown in Fig. 1.1.

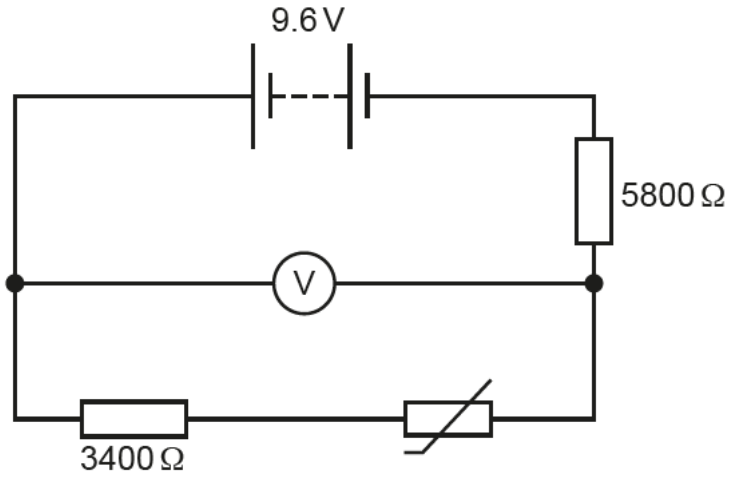


Fig. 1.1

The fixed resistors have resistances of 3400 Ω and 5800 Ω. The reading on the voltmeter in the circuit is 6.0 V.

(a) Calculate the current in the resistor of resistance 5800 Ω.

$9.6 = 6.0 + (I \times 5800)$ or $3.6 = I \times 5800$ [C1]

$I = 6.2 \times 10^{-4}$ A [A1]

current = ... A [2]

(b) Calculate the resistance of the thermistor.

$9.6 = 6.2 \times 10^{-4} \times (3400 + 5800 + R)$ [C1]

or

$6.0 = 6.2 \times 10^{-4} \times (3400 + R)$

$R = 6.3 \times 10^{3}\ \Omega$ [A1]

resistance = ... Ω [2]

(c) The initial energy stored in the battery is 2.6×10^4 J.

Assume that the e.m.f. of the battery is constant.

Determine the final energy stored in the battery after a charge of 330 C has moved through it.

$(\Delta W = \varepsilon \times Q =) 9.6 \times 330 (= 3170 \text{ J})$ [C1]

final stored energy = $2.6 \times 10^4 - 3170$
$= 2.3 \times 10^4$ J [A1]

final stored energy = ... J [2]

(d) The environmental conditions change causing an increase in the resistance of the thermistor.

State whether there is a decrease, increase or no change to:

(i) the temperature of the thermistor

decrease [B1]
(since an NTC thermistor increases in resistance when temperature falls) . [1]

(ii) the current in the thermistor

decrease [B1]
. (since total resistance increases with the e.m.f. fixed) ... [1]

(iii) the potential difference across the thermistor.

increase [B1]
... (p.d. across the fixed resistors decreases, and since the total e.m.f. is fixed, the p.d. across the thermistor increases) ... [1]

(e) If the temperature of the thermistor in Fig. 1.1 changes at a constant rate, state a reason why the potential difference across the thermistor does not change at a constant rate.

.. change of resistance with temperature is not linear (Thermistor is a non-ohmic conductor) [B1] .....

OR

.. potential divider equation is non-linear .....

..... [1]

[Total: 10]

2 The Earth has a mass M of 5.97×10^{24} kg. A satellite of mass m orbits the Earth in a circular geostationary orbit at a distance d above the Earth's surface.

(a) Show that the radius of the Earth, R, is approximately 6370 km.

$$g = \frac{GM}{R^2}$$

$$\Rightarrow R = \sqrt{\frac{GM}{g}} = \sqrt{\frac{\left(6.67 \times 10^{-11}\right)\left(5.97 \times 10^{24}\right)}{9.81}} \quad \text{[B1]}$$

$$= 6.37 \times 10^{6} \text{ m} = 6370 \text{ km} \quad \text{[A0]}$$

[1]

(b) By considering the orbit of the geostationary satellite, show that the distance d above the Earth's suface is 35900 km.

gravitational force acting on satellite by the Earth
provides the centripetal force on satellite:

$$\Rightarrow \frac{GMm}{r^2} = \frac{mv^2}{r} = \frac{m(r\omega)^2}{r} = \frac{m\left(r\frac{2\pi}{T}\right)^2}{r} \quad \text{[B1]}$$

$$\Rightarrow T^2 = \frac{4\pi^2 r^3}{GM} \Rightarrow r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$

$$\Rightarrow r = \sqrt[3]{\frac{\left(6.67 \times 10^{-11}\right)\left(5.97 \times 10^{24}\right)(24 \times 60 \times 60)^2}{4\pi^2}} = 42226.9 \text{ km} \quad \text{[B1]}$$

$$d = 42226.9 - 6370 = 35856.9 \text{ km} = 35900 \text{ km (to 3 s.f.)} \quad \text{[A0]}$$

[2]

(c) For the satellite of mass m that is in a circular orbit of radius r around the Earth of mass M, show that its total mechanical energy E can be expressed as $E = -\frac{GMm}{2r}$.

gravitational force provides the centripetal force:

$$\Rightarrow \frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow E_k = \frac{1}{2}mv^2 = \frac{GMm}{2r} \quad \text{[B1]}$$

$$E = E_k + E_p$$

$$E = \frac{GMm}{2r} + \left(-\frac{GMm}{r}\right) \quad \text{[B1]}$$

$$E = -\frac{GMm}{2r} \quad \text{[A0]}$$

[2]

(e) The satellite has a mass of 490 kg and carries 360 kg of lithium battery pack with energy density of 324 Wh kg^{-1}, of which 65% of its energy is converted to useful energy for orbital manoeuvres.

(i) Show that the amount of useful energy is 2.73×10^8 J.

useful energy = $360 \times 324 \times 60 \times 60 \times 0.65 = 2.73 \times 10^8$ J [A1]

[1]

(ii) Calculate the maximum orbital radius r_{max} the satellite can reach from its geostationary orbit.

$$\Delta E = E_f - E_i = -\frac{GMm}{2}\left(\frac{1}{r_{max}} - \frac{1}{r_i}\right)$$

$$2.73\times10^8 = -\frac{\left(6.67\times10^{-11}\right)\left(5.97\times10^{24}\right)(850)}{2}\left(\frac{1}{r_{max}} - \frac{1}{4.22\times10^7}\right) \quad \text{[M1]}$$

$$\frac{1}{r_{max}} - \frac{1}{4.22\times10^7} = -1.613\times10^{-9}$$

$r_{max} = 45300$ km [A1]

r_{max} = ... m [2]

[Total: 8]

3 A fixed mass of gas undergoes a cycle of changes of pressure, volume and temperature as shown in Fig. 3.1. The gas is assumed to be ideal.

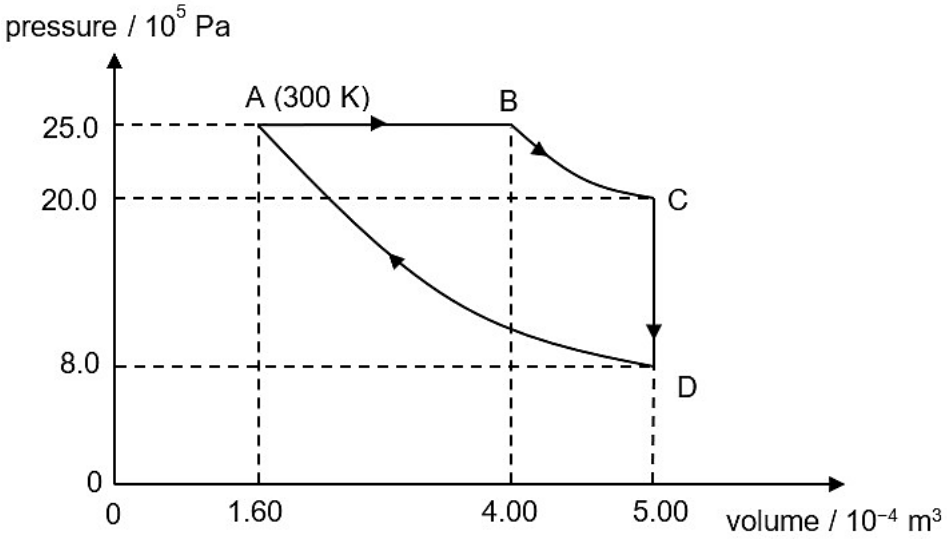


Fig. 3.1 (Not to scale)

The temperature of the gas at A is 300 K.

(a) Show that the amount of gas is 0.160 mol.

$$pV = nRT \Rightarrow n = \frac{pV}{RT} = \frac{(25.0\times10^5)(1.60\times10^{-4})}{(8.31)(300)} \quad \text{[M1]}$$

$$= 0.160 \quad \text{[A0]}$$

[1]

(b) Calculate the temperature at B.

$$pV = nRT \Rightarrow n = \frac{pV}{RT}$$

$$0.16 = \frac{(25.0\times10^5)(4.00\times10^{-4})}{(8.31)(T)} \quad \text{[M1]}$$

$$T = 750 \text{ K} \quad \text{[A0]}$$

temperature = ... K [1]

Fig. 3.2 indicates the type of the various processes in the cycle:

Process	Type of process
A to B	isobaric (constant pressure)
B to C	isothermal (constant temperature)
C to D	isochoric (constant volume)

Fig. 3.2

(c) Explain why process D to A is likely an isothermal process.

from $pV = nRT$. Since n and R are constants, the product pV must remain constant for temperature to be constant for isothermal process. [B1]

Checking the endpoints: at D, $pV = 8.0 \times 10^5 \times 5.00 \times 10^{-4} = 400$ J. At A, $pV = 25.0 \times 10^5 \times 1.60 \times 10^{-4} = 400$ J. [B1]

Since pV is the same at both points D and A, the temperature is unchanged between D and A. Hence the process D to A is likely isothermal.

..... [2]

Fig. 3.3 lists some data about the gas:

Quantity	Value
molar mass	0.0200 kg mol^{-1}
energy transferred to the gas from A to B by heating	1500 J
specific heat capacity at constant volume	625 J kg^{-1} K^{-1}

Fig. 3.3

(d) (i) Using the data from Fig. 3.3, show that the specific heat capacity of the gas from A to B at constant pressure is 1040 J kg^{-1} K^{-1}.

For process A to B:

$q = mc\Delta T = (0.0200)(0.160)c(750 - 300) = 1500$ J [B1]

$\therefore c = 1040$ J kg^{-1} K^{-1}

[1]

(ii) Use the first law of thermodynamics to determine the increase in internal energy ΔU of the gas from A to B.

$\Delta U = q + w$

$= 1500 + \left[-\left(25.0 \times 10^5\right)\left(2.40 \times 10^{-4}\right)\right]$ [M1]

$= 900$ J [A1]

Do not accept if formula for internal energy was used to determine ΔU due to question requirement.

ΔU = ... J [2]

(iii) Using the data from Fig. 3.3, calculate the heat flow during the process C to D.

process C to D:

$$q = mc\Delta T$$
$$= (0.0200)(0.160)(625)(300 - 750)$$
$$= -900 \text{ J}$$ [A1]

heat flow during the process C to D = ... J [1]

(e) Fig. 3.4 shows the increase in internal energy ΔU of the gas, the heat supplied q to the gas, and the work done on the gas w during the various processes of the cycle. During B to C, 220 J of energy is transferred to the gas by heating and during D to A, work done on the gas is 320 J.

Process	ΔU / J	q / J	w / J
A to B	**900**	**1500**	**−600**
B to C	**0**	**220**	**−220**
C to D	**−900**	**−900**	**0**
D to A	**0**	**−320**	**320**

Fig. 3.4 (1 mark for each correct row.)

Complete the table in Fig. 3.4. [4]

[Total: 12]

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4 A non-uniform L-shaped beam of weight 10 N is attached to a wall using a hinge. The longer part of the beam is 50 cm and the shorter part of the beam is 10 cm. The weight of the beam acts at a point on the beam that is at a horizonal distance of *x* away from the hinge. The beam is held at rest using a cord, such that the longer part of the beam is horizontal. *AO* is horizontal while *OC* is vertical. The tension in the cord is 5.0 N, at an angle of 60° from *OC*, as shown in Fig. 4.1.

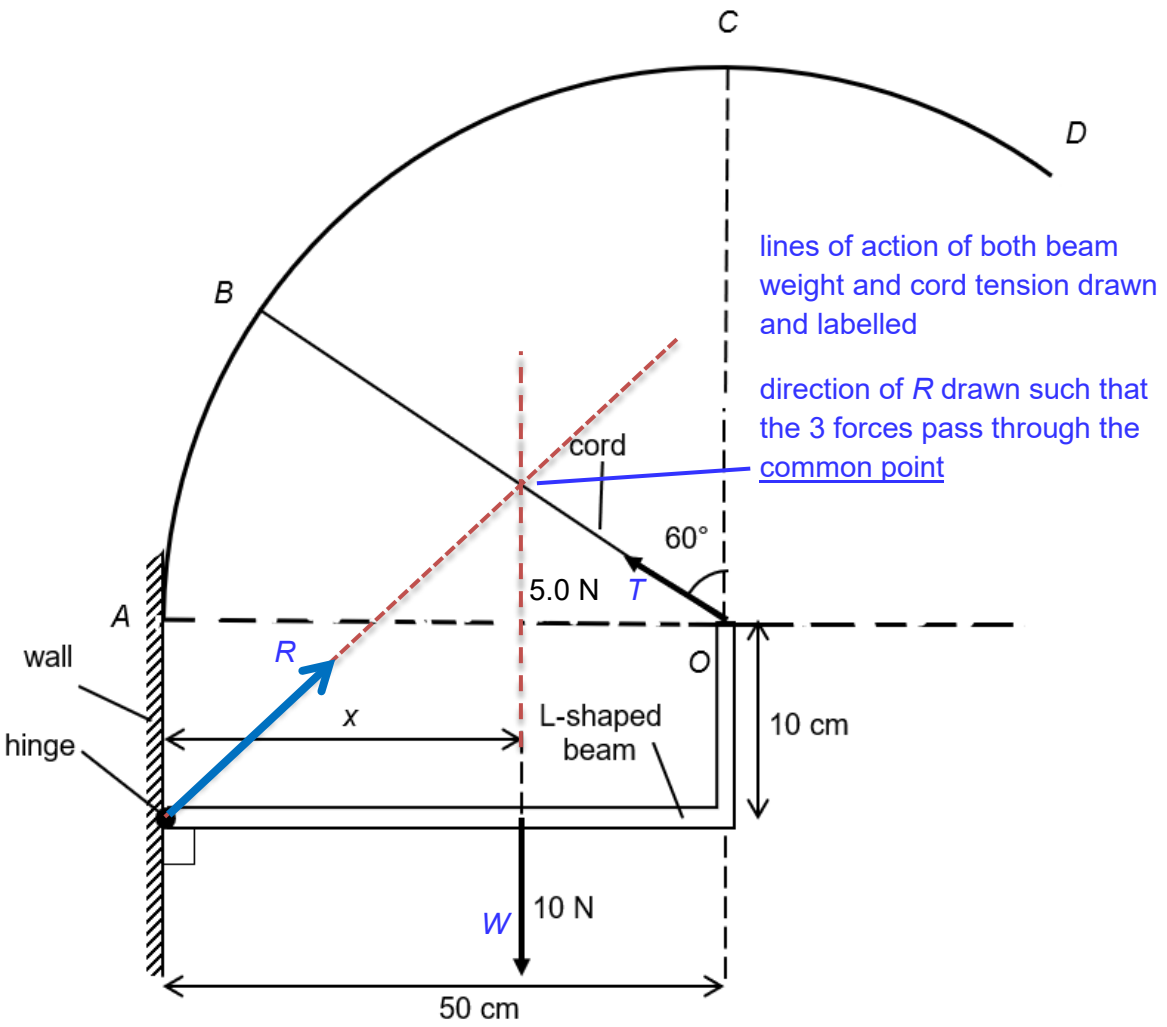


Fig. 4.1

(a) On Fig. 4.1, draw and label the contact force *R* acting on the L-shaped beam due to the hinge.

[2]

(b) By considering the forces acting on the beam, calculate the magnitude of R.

$\sum F_x = 0 \Rightarrow R_x = T_x = 5.0\sin 60^\circ = 4.33$ N

$\sum F_y = 0 \Rightarrow R_y + T_y = W \Rightarrow R_y = W - T_y = 10 - 5.0\cos 60^\circ = 7.50$ N

} [M1]

$|R| = \sqrt{R_x^2 + R_y^2} = \sqrt{4.33^2 + (-7.50)^2}$ [M1]

$= 8.66$ N

$= 8.7$ N (to 2 s.f.) [A1]

magnitude of R = ... N [3]

(c) Determine the value of x.

using hinge as pivot:
clockwise moment due to weight = anticlockwise moment due to tension

$Wx = \left[T\left(\cos 60^\circ\right)\right](0.50) + \left[T\left(\sin 60^\circ\right)\right](0.10)$

$10x = 5.0\left[0.50\left(\cos 60^\circ\right) + 0.10\left(\sin 60^\circ\right)\right]$ [M1]

$x = 0.168$ m $= 0.17$ m $= 17$ cm [A1]

x = ... cm [2]

(d) *ABCD* are different points on a circle where the centre is at point *O*. *C* is at the point vertically above point *O*. The cord moves along the arc of the circle such that the beam remains stationary.

(i) State how the magnitude of the contact force *R* would change as the cord moves from *B* to *C*.

Magnitude *R* decreases as the cord moves from *B* to *C*. [B1]

..... [1]

(ii) State how the direction of the contact force *R* would change as the cord moves from *C* to *D*. Mathematical calculation is not required in this part.

At *C*, *R* acts vertically upwards. As the cord moves from *C* to *D*, the direction of *R* rotates anticlockwise into the wall. [B1]

..... [1]

[Total: 9]

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5 Two point charges, A and B, are separated in a vacuum by a distance of 0.0100 cm, as shown in Fig. 5.1.

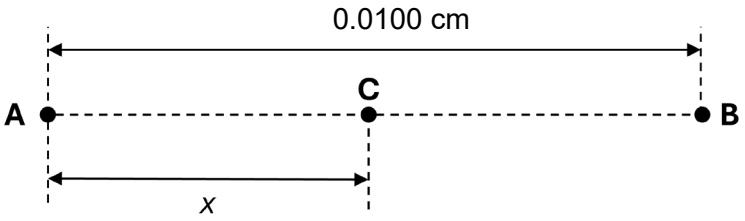


Fig. 5.1

A point C is on the line joining the two charges and is a distance x from charge A.

The variation with distance x of the electric potential V_x at point C is shown in Fig. 5.2.

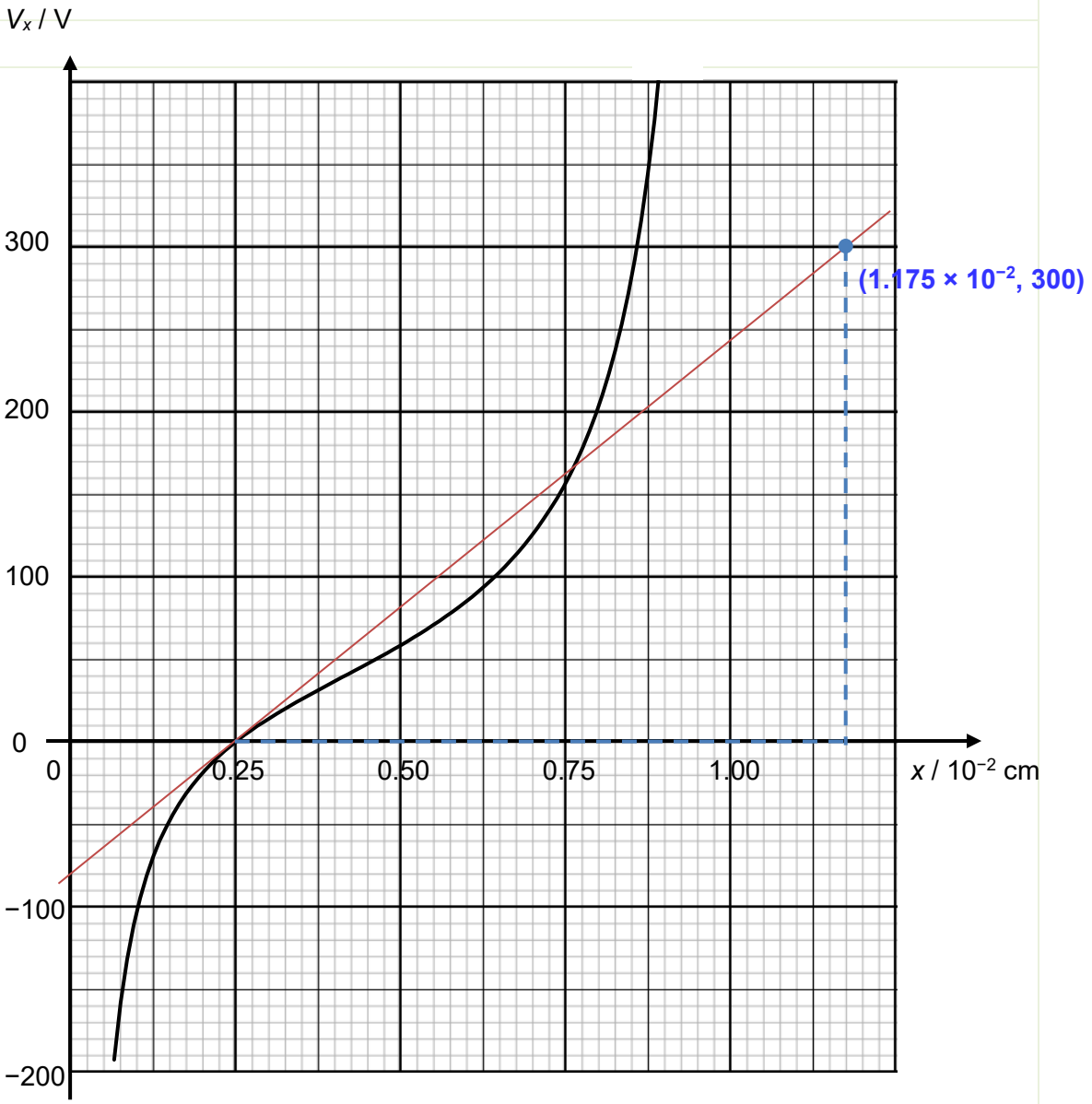


Fig. 5.2

(a) Define electric field strength at a point.

Electric field strength at a point in an electric field is defined as the electric force per unit positive charge exerted on a small test charge placed at that point. [B1]

..... [1]

(b) Explain whether the charges have the same or opposite signs.

The graph shows that the electric potential is zero at a point between the two charges (at x = 0.0025 cm), which is only possible if the two charges are of opposite signs — a positive charge produces a positive potential whilst a negative charge produces a negative potential. [2]

OR

Furthermore, since electric field strength is equal to the negative potential gradient, and the potential decreases continuously from B to A, the resultant electric field points from B towards A at every point between the charges. For this to be the case, the fields due to each individual charge must reinforce rather than cancel, which is only possible if the charges are of opposite signs. [2]

..... [2]

(c) Calculate the electric field strength where $x = 0.25 \times 10^{-2}$ cm.

$$\frac{dV}{dx} = \frac{(300-0)\text{ V}}{(1.175-0)\times10^{-4}\text{ m}} = 2.6\times10^{6}\text{ V m}^{-1} \quad \text{[M1]}$$

$$E = -\frac{dV}{dx}$$

$$= -2.6\times10^{6}\text{ V m}^{-1} \quad \text{[A1]}$$

Negative sign indicates E points in the direction of decreasing V. Therefore, the E has magnitude 2.6×10^{6} V m^{-1} and points in the negative x-direction (leftwards, towards charge A). [B1]

electric field strength = ... V m^{-1}

direction of electric field strength: ..... [3]

(d) An electron is projected with a speed of 7.73×10^6 m s^{-1} from the position where $x = 0.725 \times 10^{-2}$ cm towards point charge A.

Calculate the speed of the electron at the position where $x = 0.175 \times 10^{-2}$ cm.

$\Delta E_k = -\Delta E_p$ {conservation of mechanical energy}

$\Delta E_p = q(V_f - V_i)$ [M1]

$= e[(-30) - (140)]$

$= (-1.60 \times 10^{-19})(-170) = 2.72 \times 10^{-17}$ J

$\therefore \Delta E_k = \frac{1}{2} m_e [v_f^2 - v_i^2] = -2.72 \times 10^{-17}$ J [M1]

$\Rightarrow v_f = \sqrt{\frac{2\Delta E_k}{m_e} + v_i^2} = \sqrt{\frac{2(-2.72 \times 10^{-17})}{9.11 \times 10^{-31}} + (7.73 \times 10^6)^2}$ **=0.0 ms^{-1}** [A1]

speed = ... m s^{-1} [3]

[Total: 9]

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6 A simple ideal iron-cored transformer is illustrated in Fig. 6.1.
An alternating input p.d. V_p of 240 V is connected across the primary coil, of 120 turns of the transformer. The secondary p.d. V_s of 3600 V is applied to a load resistor.

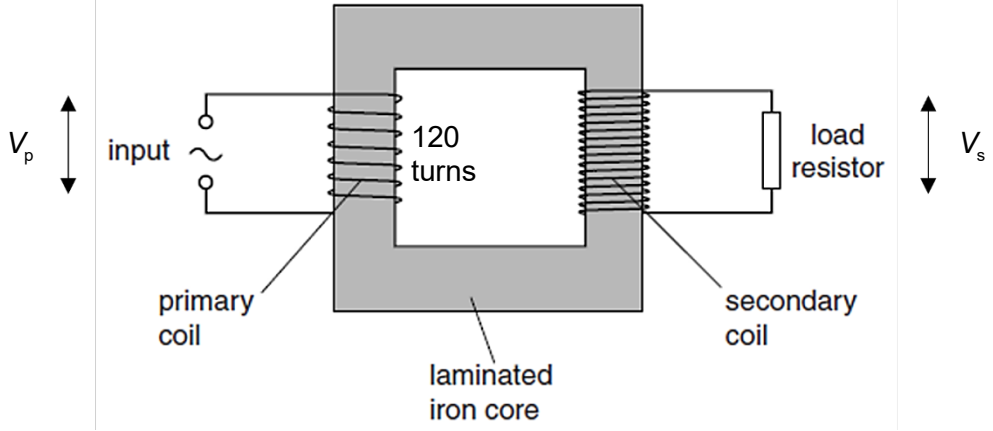


Fig. 6.1

(a) (i) Explain why an iron core is preferred over an air core in the construction of a transformer.

. it increases the magnetic flux density OR concentrates the magnetic flux as it links it from the primary to the secondary coil, (thereby ……
. increase the efficiency of energy transfer) [B1] ……

…… [1]

(ii) Explain why the iron core is laminated.

. to reduce energy losses by reducing eddy (induced) currents [B1] …….

……

…… [1]

(b) Calculate the number of turns in the secondary coil.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} \Rightarrow N_s = 120\left(\frac{3600}{240}\right) = 1800 \quad \text{[B1]}$$

number of turns = ... [1]

	(c)	Using Faraday's Law, explain why an alternating current in the primary coil gives rise to an alternating e.m.f. in the secondary coil.
		The alternating current in the primary coil produces a **magnetic flux that changes continuously** in both magnitude and direction. [B1] Since the secondary coil is wound on the same iron core, **this changing magnetic flux links the secondary coil**, so the flux through the secondary coil also alternates. [B1] By Faraday's Law, the induced e.m.f. is proportional to the rate of change of magnetic flux linkage [$\varepsilon = -N(d\Phi/dt)$]. Since the flux is continuously changing, an e.m.f. is continuously induced in the secondary coil. **As the flux alternates, the rate of change also alternates in sign,** so the induced e.m.f. alternates as well. [B1] …… …… …… …… …… [3]
	(d)	A step-up transformer is used when transmitting electricity over long distances. Suggest a reason for this.
		…… A step-up transformer increases the voltage and reduces the current, thereby reducing the power loss in the transmission cables ($P = I^2R$), making transmission more efficient. [B1] …… …… …… …… [1]
		[Total: 7]

Section B

Answer **one** question in this section in the spaces provided.

7 Fig. 7.1 shows a small mass attached to the end of a fixed spring.

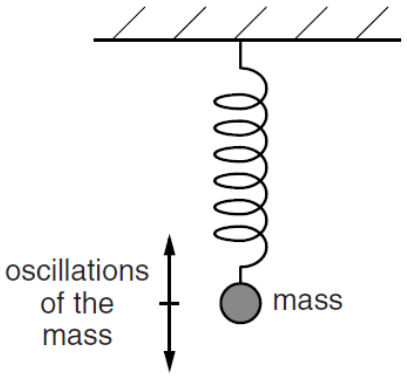


Fig. 7.1 (not to scale)

The mass is given a small vertical displacement and then released. It oscillates with simple harmonic motion about its equilibrium position.

(a) Define the term *simple harmonic motion*.

acceleration is proportional to displacement from fixed point [1]

acceleration directed to fixed point (equilibrium point) [1]

.

.....

..... [2]

(b) The graph in Fig. 7.2 shows how the displacement x varies with time t for the motion of the mass on the spring.

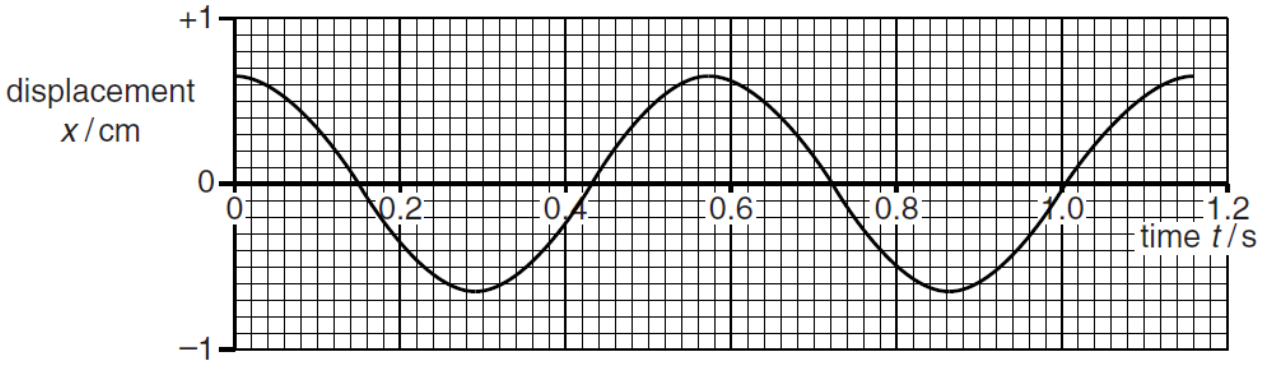


Fig. 7.2

(i) Using data from the graph in Fig. 7.2, determine the velocity of the mass at time t = 0.66 s.

(from graph in Fig. 7.2)

T = 0.58 (s)

A = 0.65 (cm) [1]

$\omega = 2\pi/T$ [1]

substitution $v = -A\omega \sin \omega t = -0.65 \times 10.83 \sin (10.83 \times 0.66)$ [1]

$= -5.4$ cm s^{-1}

OR

the gradient of tangent at $t = 0.66$ s is negative, so

$v = -\omega\sqrt{x_0^2 - x^2} = -\dfrac{2\pi}{0.58}\sqrt{0.65^2 - 0.40^2} = -5.6$ cm s^{-1}

velocity = ... cm s^{-1} [3]

(iii) Fig. 7.3 shows the spring and the position of the mass at instant t = 0.94 s.

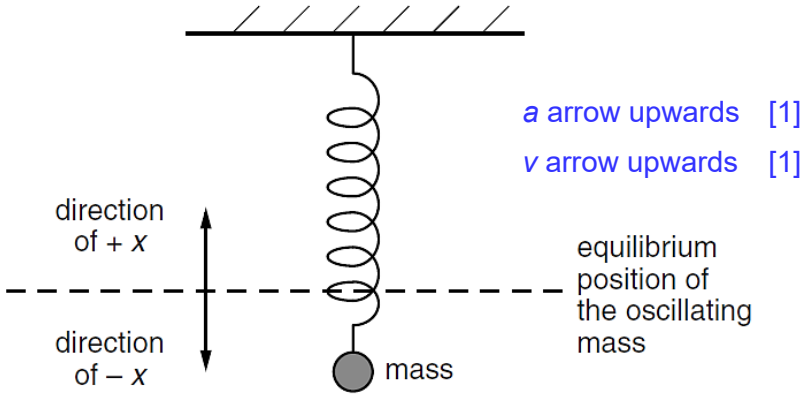


Fig. 7.3

On Fig. 7.3, add arrows labelled *a* and *v* to show the directions of the acceleration and the velocity, respectively, at this instant in time. [2]

a arrow upwards [1]

v arrow upwards [1]

(c) Fig. 7.3 shows a uniform rod floating vertically in water of density ρ. The rod has a mass M and a uniform cross-sectional area A.

When the rod is in equilibrium, its base is at a depth h below the water surface.

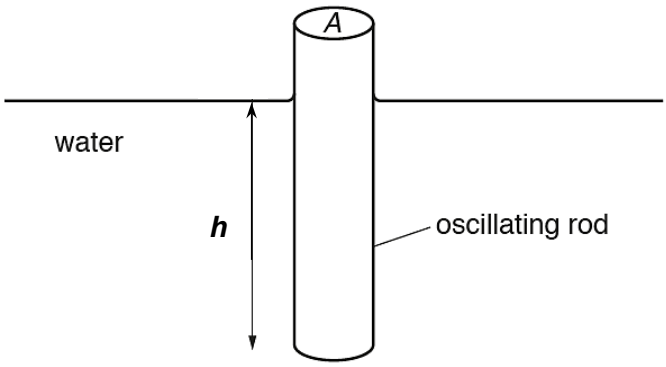


Fig. 7.3

The rod is displaced vertically and released. The acceleration a of the resulting oscillations is given by

$$a = -\left(\frac{\rho A g}{M}\right) x$$

where x is the displacement from the equilibrium depth h and g is the acceleration due to free fall.

(i) Show that the motion of the rod is simple harmonic.

The given equation is:

$$\boldsymbol{a = -\left(\frac{\rho A g}{M}\right) x}$$

Since ρ, M, A and g are constants, the acceleration is directly proportional to displacement x and is in opposite direction to x (indicated by the negative sign), the motion of the rod is simple harmonic. [1]

---

the motion is simple harmonic, with:

$$\boldsymbol{\omega^2 = \frac{\rho A g}{M}}$$

[1]

[2]

(ii) Hence, derive an expression for the period T of the oscillations, in terms of ρ, M, A and g.

$\omega^2 = \left(\frac{\rho A g}{M}\right)$ [M1]

$\Rightarrow T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{M}{A\rho g}}$ [A1]

[2]

(iii) The rod is given a small vertical displacement and then released. It oscillates, but damping soon causes it to come to rest after a few oscillations in the water.

State why the damping causes the oscillations to stop.

· (friction / drag acts on the rod and) loss of energy (causes amplitude decrease) [1]

.....

.....[1]

(iv) Information from Fig. 7.4 can be used to verify that the damping on the rod is exponential.

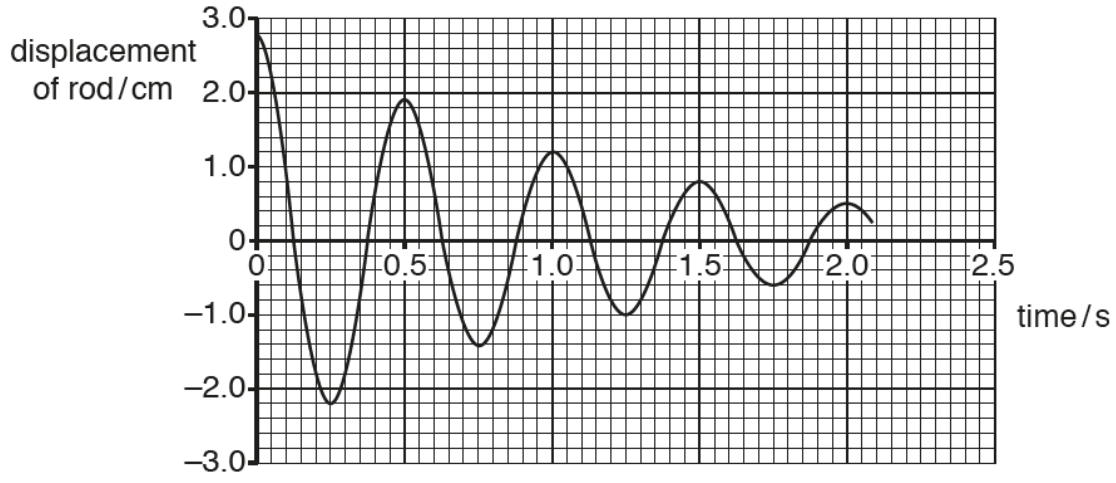


Fig. 7.4

1. Use Fig. 7.4 to complete the second column in the table of Fig. 7.5 to record the displacement at the times stated. [1]

2. Use the displacement data and the blank column in Fig. 7.5 to show that the rod exhibits light damping, where its amplitude decays exponentially with time. [1]

3. Use the grid in Fig. 7.6 to draw a graph that illustrates your answer. [2]

time / s	displacement / cm	
0.0		
0.5		
1.0		
1.5		
2.0		

Fig. 7.5

1. 5 results correct [1]

only 1 error is 1 mark only

more than 1 error 0 marks

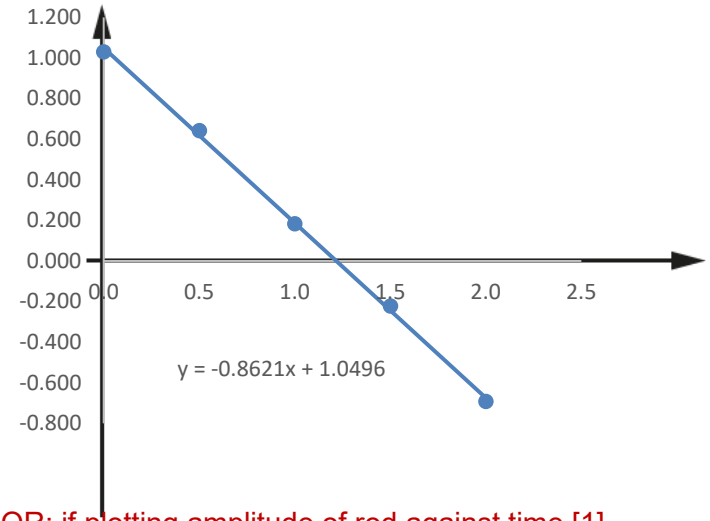
time / s	displacement / cm
0.0	2.8
0.5	1.9
1.0	1.2
1.5	0.8
2.0	0.5

2. calculate the values of [ln (displacement)] and show constant difference OR calculate the ratios of consecutive values and show constant value [1]

time / s	displacement / cm	ln displacement
0.0	2.8	1.03
0.5	1.9	0.642
1.0	1.2	0.182
1.5	0.8	–0.223
2.0	0.5	–0.692

3. [ln (displacement)] against (time) fully labelled [1]

straight line graph of negative gradient [1]



OR: if plotting amplitude of rod against time [1],
show that the half-lives are the same from the graph [1]

Fig. 7.6

(d) Earthquake waves can produce forced oscillations in tall buildings. The buildings may collapse because resonance occurs.

Explain what is meant by the term *resonance* in this situation.

In this situation, the earthquake acts as the driving force and the building acts as the oscillating system. Resonance occurs when the frequency of the driver matches the natural frequency of the system [1], resulting in a maximum transfer of energy and a maximum amplitude of oscillation. [1]

(OR) Resonance occurs when the frequency of the external driver (the earthquake) matches the natural frequency of the oscillating system (the building) [1]. This leads to a maximum transfer of energy from the driver to the system, causing the building to oscillate with maximum amplitude. [1]

..... [2]

(e) Fig. 7.7 shows a tall building designed to prevent collapse during an earthquake. The building has a solid base which rests on a rubber bearing.

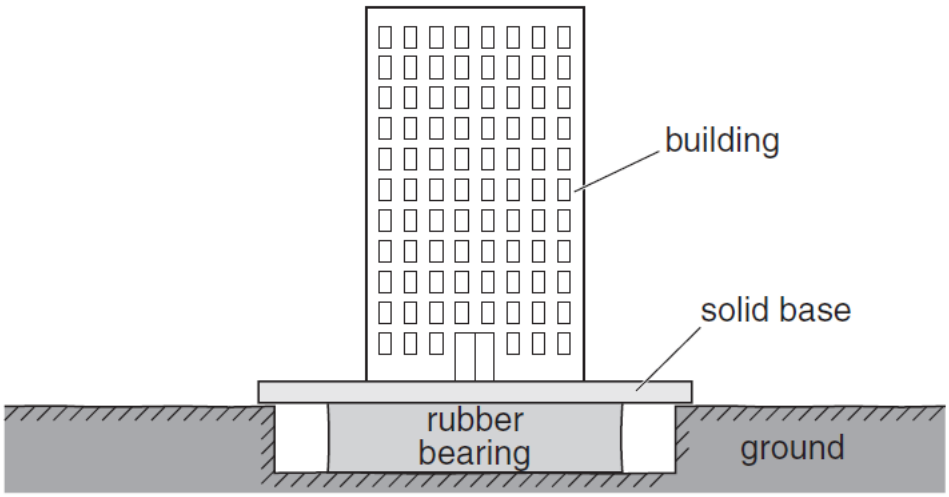


Fig. 7.7

Suggest why this design helps to prevent the building from collapse.

.....

The rubber bearing absorbs/dissipates kinetic energy from the earthquake (transforming it into thermal energy). [1]
OR: The rubber bearing increases the degree of damping of the system (the building).
Accept: Acts as a damper / absorber to absorb vibrations.

This reduces the maximum amplitude of the building's oscillations (at resonance). [1]
OR: It flattens the resonance curve, keeping the amplitude within safe limits.
Accept: Reduces the size/severity of the swings.

OR:

The rubber bearing increases the degree of damping within the oscillating system (the building) by absorbing and dissipating kinetic energy from the earthquake waves. [1]

In terms of a resonance graph, this heavier damping flattens the resonance peak, thereby drastically decreasing the maximum amplitude of the building's oscillations and preventing collapse. [1]

..... [2]

[Total: 20]

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8 **(a)** A beam of electrons is directed along a normal towards a barrier, as shown in Fig. 8.1.

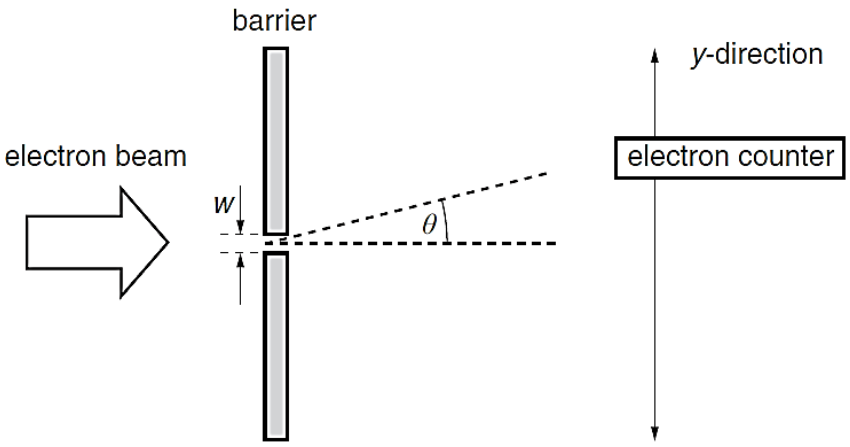


Fig. 8.1

The barrier contains a single slit of width w. Beyond the slit there is a counter that counts electrons. This can be moved in the y-direction to compare the rate of arrival of electrons at different values of the angle θ from the original direction of the beam.

(i) Louis de Broglie suggested that particles such as electrons have an associated wavelength. State the equation for the de Broglie wavelength of an electron and identify each term.

.....

$\lambda = h/p$,

where λ is the de Broglie wavelength, h is the Planck constant, and p is the momentum of the electron. [1]

.....

(ii) State the condition for diffraction to be observable in this scenario.

The de Broglie wavelength of the electrons is comparable to the slit width w. [1]

.....

(iii) Electrons are accelerated by an accelerating voltage. State and explain what happens to angle θ when magnitude of the accelerating voltage decreases.

When the accelerating voltage is decreased, the electrons (loses less electric potential energy and) gains less kinetic energy; hence, its speed v is decreased and momentum, (the product of its mass m and v) decreases. Since de Broglie's wavelength is inversely proportional to its momentum, wavelength increases. [1]

The electrons will be more diffracted ($d \sin\theta = n\lambda$) and hence the angle θ increases. [1]

[2]

(b) Werner Heisenberg used a different approach involving what is now known as the uncertainty principle. This can also be used to explain why the electron beam spreads out after passing through the slit. One version of this involves the equation

$$\Delta y \Delta p \gtrsim h \, .$$

Using the uncertainty principle, explain what happens to angle θ when the slit is made narrower.

A narrower slit means a smaller value of w, so Δy is smaller. [1]

By the uncertainty principle $\Delta y \Delta p \gtrsim h$, a smaller Δy would mean a larger Δp, meaning the uncertainty in the (y-component of) momentum is greater. [1]

A larger spread in y-momentum means electrons travel across a wider angles θ, so the beam is spread out more. [1]

[3]

(c) Explain whether your answer in **(b)** is consistent with single slit diffraction, when the slit is made narrower.

In single slit diffraction, $\sin\theta = \lambda/b$, the angle θ is inversely proportional to slit width, when wavelength is constant. So, it results in a larger angle θ. [1]

By the uncertainty principle, a narrower slit implies a larger spread in y-momentum means electrons travel across a wider range of angles, so the beam is spread out more it results in a larger angle θ. [1]

The answer is consistent with single slit diffraction.

[2]

(d) Fig. 8.2 shows a cooler region of hydrogen gas surrounding a hot gas cloud emitting white light.

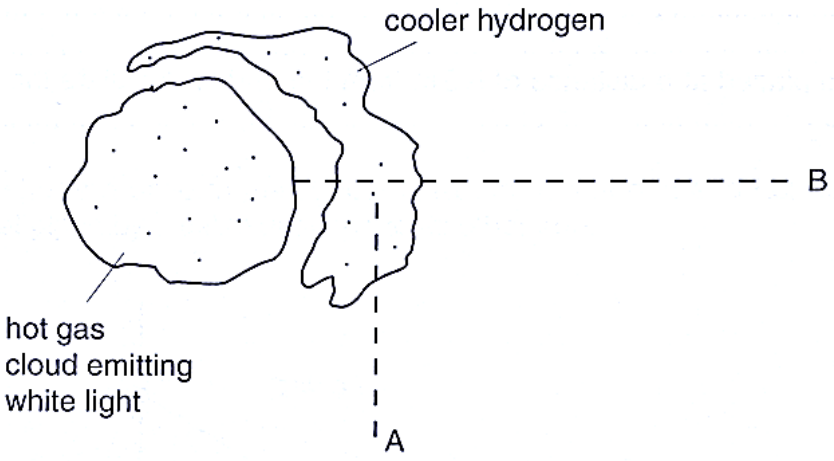


Fig. 8.2

		(i)	Describe the difference in appearance between an emission and absorption line spectra.
			…… …. Emission line spectra consist of discrete bright lines of different colours on a dark background … …. Absorption line spectra are continuous spectrum crossed by dark lines. … [1] …… …… …… [1]
		(ii)	State the types of spectra seen from point A and B.
			1. point A: …… A: Emission line spectra ……. **2.** point B: …… B: Absorption spectra ……. [1]
		(iii)	Explain how spectra seen at point B is obtained. Energy levels in gas atoms are discrete. [1] When white light passes through the cooler hydrogen, only photons that has energy equal to the difference between two energy levels in the gas atom will be absorbed to cause electron excitation. [1] The unabsorbed photons will form a continuous spectrum of light on the screen with discrete dark lines. Electrons in the excited state are unstable and will de-excite to ground state to produce photons with energy equal to the difference between the excited state and lower state (accept ground state). These photons are emitted in random direction and will be missing from the spectra at point B. [1] …… …… …… [3]

(e) The energy levels E_n in a hydrogen atom are given by the empirical equation

$$E_n = \frac{-13.6\ \text{eV}}{n^2},$$

where n is an integer.

The four visible spectral lines P, Q, R and S in the hydrogen spectrum are illustrated in Fig. 8.3.

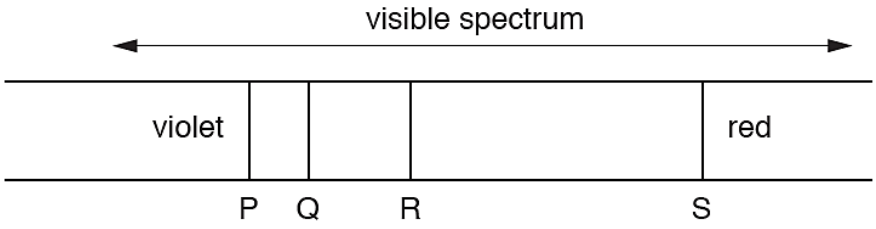


Fig. 8.3

The frequencies of three of these spectral lines are shown in the table.

spectral line	frequency / 10^{12} Hz
P	731
Q	f_Q
R	617
S	456

These frequencies are emitted when the energy level to which the electrons arrive is for $n = 2$.

(i) Deduce the value of n for the energy level from which an electron has moved in order to produce spectral line S.

$$E_n - E_2 = hf$$

$$\left(\frac{-13.6\ \text{eV}}{n^2}\right) - \left(\frac{-13.6\ \text{eV}}{2^2}\right) = \left(6.63\times10^{-34}\right)\left(456\times10^{12}\right) \quad [1]$$

$$\left(-13.6\times1.60\times10^{-19}\ \text{J}\right)\left(\frac{1}{n^2} - \frac{1}{4}\right) = 3.0233\times10^{-19} \quad [1]$$

$$\left(\frac{1}{n^2} - \frac{1}{4}\right) = -0.139$$

$$n = 3 \quad [1]$$

n = ... [3]

		(ii)	Determine the frequency of the spectral line Q, f_Q. $E_5 - E_2 = hf_Q$ [1] $\left(\frac{-13.6\text{ eV}}{5^2}\right) - \left(\frac{-13.6\text{ eV}}{2^2}\right) = \left(6.63\times10^{-34}\right)f_Q$ [1] $\left(-13.6\times1.60\times10^{-19}\text{ J}\right)\left(\frac{1}{25} - \frac{1}{4}\right) = \left(6.63\times10^{-34}\right)f_Q$ $f_Q = 689\times10^{12}\text{ Hz}$ [1] f_Q = ... Hz [3]
			[Total: 20]

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